

Camille Robichon

R&D Formulation Scientist – Lipid Systems & Antioxidants

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Research Profile

Formulation scientist specialised in lipid-based delivery systems, emulsions, liposomes, niosomes and antioxidant strategies. Experienced in translating molecular mechanisms into innovative lipid formulations with improved oxidative stability and bioactive delivery. Skilled in experimental design, kinetic modelling, and multidisciplinary analytical techniques for food, nutraceutical and cosmetic applications. International research experience in France, the United Kingdom, and Japan.

Education

- 2023–Present **PhD in Food Science**, *University of Montpellier*, Montpellier, France
Expected completion: October 2026
Thesis: *New Formulation Strategies for Vitamin E to Improve Antioxidant Efficiency in Food Matrices.*
- 2020–2023 **Chemical Engineering Degree**, *École Nationale Supérieure de Chimie de Lille (ENSCL)*, Lille, France
Specialisation: *Formulation Chemistry.*
- Sep 2022–Mar 2023 **Exchange Semester**, *Doshisha University*, Kyoto, Japan
Project: Transdermal delivery of curcumin using an oil-in-water microemulsion.

Research Experience

- Nov 2023–Present **PhD Candidate in Food Science**, *CIRAD, UMR QualiSud*, Montpellier, France
New Formulation Strategies for Vitamin E to Improve Antioxidant Efficiency in Food Matrices.
- Designed lipid formulations and screened antioxidant combinations to improve vitamin E stability and oxidative resistance in oil-in-water emulsions.
 - Developed and compared Weibull, Gompertz, Logistic and Exponential models to characterise lipid oxidation kinetics and quantify antioxidant efficiency and interaction effects.
 - Developed and optimised lipid model systems and nanoemulsions using microfluidisation to investigate oxidation mechanisms.
 - Formulated and characterised liposomes and niosomes using DLS and HPLC and evaluated their effects on vitamin E stability and bioaccessibility.
 - Monitored lipid oxidation kinetics using NMR, HPLC, GC–MS, and UV–Vis spectroscopy.
 - Investigated the distribution of bioactive compounds between emulsion droplets and surfactant micelles using Taylor dispersion analysis.
 - Evaluated vitamin E bioaccessibility using a static *in vitro* gastrointestinal digestion model.
 - Research outcomes include two first-author publications and one manuscript in preparation.
- 2023 (6 months) **End-of-Studies Research Project**, *BioWooEB Research Group, CIRAD*, Montpellier, France
Optimization of a method for extracting proteins from dry rubber samples and development of a method for quantifying proteins using IR-TF
- Performed SDS-PAGE analyses.
 - Developed an infrared spectroscopy method for the characterisation of rubber proteins.

- 2022–2023 (6 months) **Exchange Semester, Doshisha University, Kyoto, Japan**
Transdermal delivery of curcumin using an oil-in-water microemulsion based on isopropyl myristate.
- Formulated and characterised curcumin-loaded oil-in-water microemulsions.
 - Evaluated *in vitro* transdermal delivery using Franz diffusion cell experiments.
 - Performed spectrophotometric analyses to quantify curcumin permeation and release.
- 2022 (2 months) **Research Internship, ADS Research Group, University of St Andrews, St Andrews, Scotland, UK**
Effect of substrates on the selectivity of the kinetic resolution of tertiary pyrazolone alcohols using chiral isothioureia catalysis
- Synthesised pyrazolone derivatives and performed kinetic resolution using isothioureia catalysis.
 - Purified compounds using column chromatography.
 - Performed HPLC, NMR, and infrared spectroscopy analyses for structural characterisation and purity assessment.
- 2020–2023 **Research Project, École Nationale Supérieure de Chimie de Lille (ENSCL), Lille, France**
- Synthesised flame-retardant compounds for polylactide, including salen-based compounds and benzoxazines.
 - Performed reactive extrusion to manufacture PLA plates.

Publications

Peer-reviewed publications

- First-author publications **Robichon, C.**, Villeneuve, P., Bohuon, P., Baréa, B., Barouh, N., Courtois, F., Fine, F., Durand, E. (2025). *Unveiling synergistic antioxidant combinations for α -tocopherol in emulsions: A spectrophotometric–mathematical approach*. *Current Research in Food Science*, 11, 101134. doi:10.1016/j.crfs.2025.101134
- Robichon, C.**, Durand, E., Gamaethiralalage, J. G., Bohuon, P., Baréa, B., Barouh, N., Courtois, F., Fine, F., de Smet, L. C. P. M., Villeneuve, P. (2026). *Decoding Tocopherol–Polyphenol Interactions in Oil-in-Water Emulsions through Combined WIM–CAT and CV Assays*. *Current Research in Food Science*, 12, 101344. doi:10.1016/j.crfs.2026.101344
- Co-author publications Durand, E., Troncho, T., Koli, H., **Robichon, C.**, Villeneuve, P. (2026). *Focus on the Role of Mixed Micelles and Lipid Droplets in the Oxidative Stability of Oil-in-Water Emulsions Using Size Distribution Taylor Dispersion Analysis*. *Current Research in Food Science*, 12, 101369. doi:10.1016/j.crfs.2026.101369
- Smith, A. D., Westwood, M. T., Farah, A. O., Wise, H. B., Sinfield, M., **Robichon, C.**, Prindl, M. I., Cordes, D. B., Cheong, P. H.-Y. (2024). *Isothioureia-Catalysed Acylative Kinetic Resolution of Tertiary Pyrazolone Alcohols*. *Angewandte Chemie International Edition*, e202407983. doi:10.1002/anie.202407983
- Manuscript in preparation **Robichon, C.**, Durand, E., Silvestre, C., Vermorel, C., Roche, H., Troncho, T., Simon, L., Repellin, M., Dhuique-Mayer, C., Fine, F., Villeneuve, P. *Liposomes and Niosomes of α -Tocopherol: Impact on Bioaccessibility and Oxidative Stability in Oil-in-Water Emulsions*.
- Submitted Villeneuve, P., **Robichon, C.**, Dhuique-Mayer, C., Durand, E. (2026). *Encapsulation of Vitamin E and Its Impact on Stability, Delivery and Activity*.

Scientific Communications

- Oral 20th Euro Fed Lipid Congress and Expo, Leipzig, Germany, 2025. *Design of a Spectrophotometric Assay Coupled with Mathematical Modeling to Identify Optimal Antioxidant Synergies with Tocopherols in Oil-in-Water Emulsions.*
- Oral AOCS Annual Meeting & Expo, Portland, USA, 2025. *Development of a Spectrophotometric Assay with Mathematical Modeling to Unveil Optimal Combinations for Antioxidant Synergy with Tocopherols in Oil-in-Water Emulsions.*
- Poster 5th International Symposium on Lipid Oxidation and Antioxidants, Bologna, Italy, 2024. *A Rapid and Tunable Screening Method for Exploring Synergistic Antioxidant Interactions with α -Tocopherol in Oil-in-Water Emulsions.*

Software & Code

- Model comparison MATLAB framework for comparing oxidation kinetic models. Available on GitHub.
- Antioxidant modelling MATLAB scripts for modelling antioxidant efficiency. Available on GitHub.

Technical Skills

- Formulation & Systems Oil-in-water emulsions; nanoemulsions; liposomes; niosomes; microfluidisation; reactive extrusion
- In Vitro Models Static gastrointestinal digestion; vitamin E bioaccessibility; transdermal delivery; Franz diffusion cells
- Analytical Techniques HPLC; GC-MS; NMR; UV-Vis spectroscopy; infrared spectroscopy; DLS; Taylor dispersion analysis; SDS-PAGE; cyclic voltammetry
- Modelling & Data Analysis MATLAB; kinetic modelling; model comparison; GraphPad Prism; VBA; Microsoft Office 365

Languages

- French Native
- English Advanced professional proficiency – TOEIC: 855/990
- Spanish Intermediate
- Japanese Beginner – currently learning

Interests

Detective novels; cycling; walking and hiking; concerts; Japanese language and culture.